

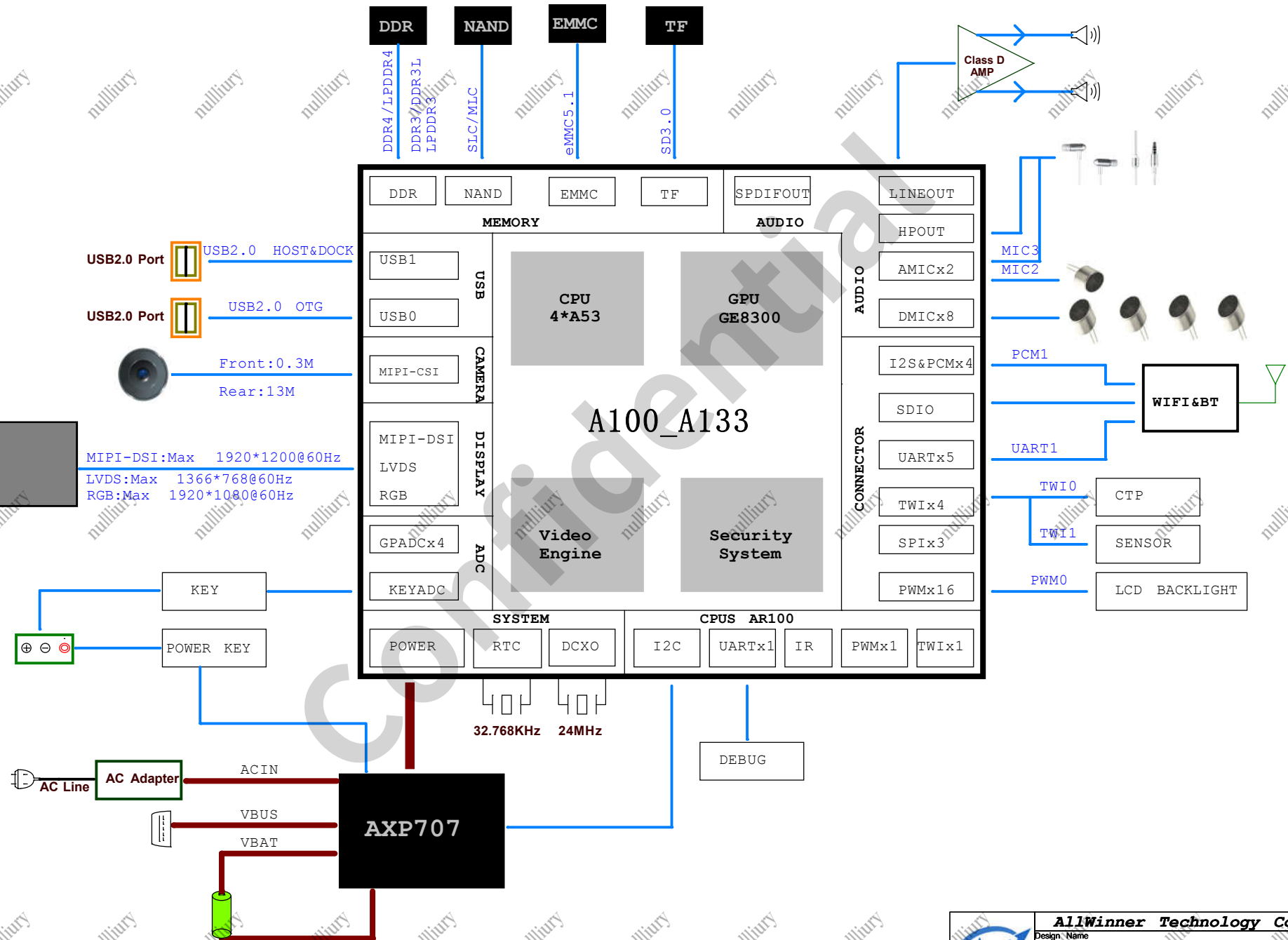
VERSION HISTORY

Index:

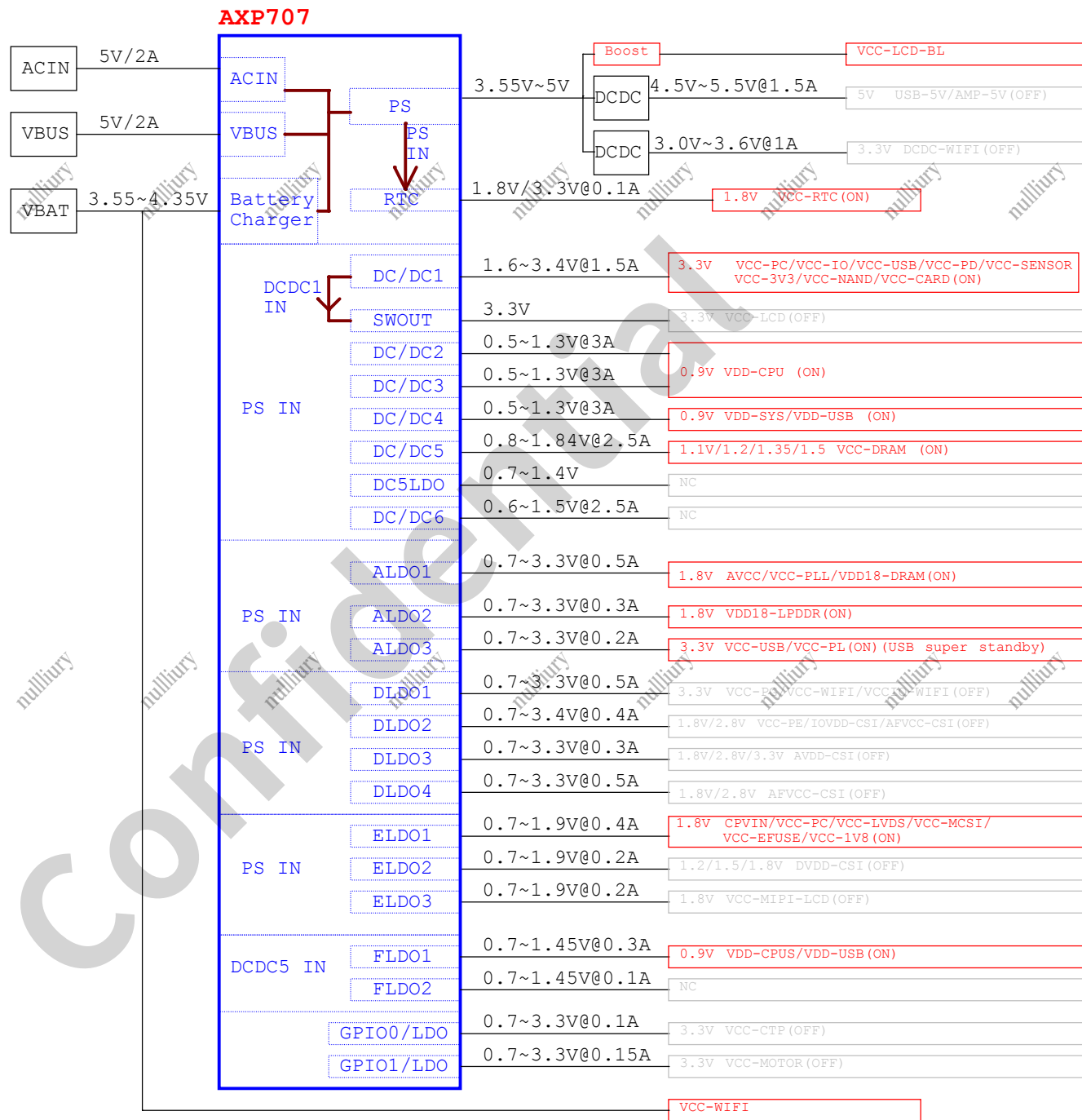
- P01 VERSION HISTORY
 - P02 BLOCK DIAGRAM
 - P03 POWER TREE
 - P04 GPIO ASSIGNMENT
 - P05 PMIC
 - P06 SOC-SYS
 - P07 SOC-GPIO
 - P08 LPDDR3 32*1
 - P09 FLASH
 - P10 AUDIO
 - P11 USB/T-CARD
 - P12 CAMREA
 - P13 MIPI LCM
 - P14 CTP
 - P15 KEY/DEBUG
 - P16 WIFI+BT
 - P17 SENSOR
- OPTION:
- P10 AUDIO+AEC
 - P13 RGB LCM
 - P16 5G WIFI+BT

Revision	Description	Date	Drawn	Checked	Approved
Ver 1.0	Release version	2020-03-12			
Ver 1.1	1.P03 page , Instructions for modifying Power Tree. 2.P07 page, Modify PL7 port function. 3.P07 page, PD port adds RGB LCM related design. 4.P10 page ,Modify the group value of power amplifier feedback resistors R22 and R24 to 100K. 5.P13 and P16 page,Increase RGB LCM and 5G WiFi+BT design options. 6.Delete AC101 design.	2020-04-30			
Ver 1.2	Update A100 symbol.	2020-05-19			
Ver 1.3	1.P05 page , Add PR45/CP83. 2.Option P13 page, Modify the RST design circuit of LCM RGB. 3.Modify A100 description as A100_A133. 4.P09 page , add BGA NAND module design. 5.Delete the description of the camera module about the I2C design.	2020-06-28			
Ver 1.4	1.P05 page , add DDR4 power supply design. 2.P07 page, LCD-BL-EN moved PB8 port to PH18. 3.Add option P10 page,add AC107 design. 4.Add 5G module external DCDC chip enable signal pull-down resistor R208.	2020-09-02			

BLOCK



☐ DEFAULT POWER ON
☐ DEFAULT POWER OFF



GPIO ASSIGNMENT

PIN	Define	CFG	Function
PB0	CPUX_TMS	4	DEBUG
PB1	CPUX_TCK	4	
PB2	CPUX_TDO	4	
PB3	CPUX_TDI	4	
PB4	I2S0_MCLK	3	I2S
PB5	I2S0_BCLK	3	
PB6	I2S0_LRCK	3	
PB7			
PB8	I2S0_DIN	3	DEBUG
PB9	CPUX_TX	2	
PB10	CPUX_RX	2	

PIN	Define	CFG	Function
PE0	MCSI_MCLK	2	CSI
PE1	MCSI_SCK	2	
PE2	MCSI_SDA	2	
PE3			
PE4			
PE5			
PE6	MCSIB_STBY_F	1	
PE7	MCSIB_RST_F	1	
PE8	MCSIA_STBY_R	1	
PE9	MCSIA_RST_R	1	

PIN	Define	CFG	Function
PH0	TWI0_SCK	2	TWI
PH1	TWI0_SDA	2	
PH2	TWI1_SCK	2	
PH3	TWI1_SDA	2	
PH4	PS-EINT	1	GPIO
PH5			
PH6	PA_SHDN	0	
PH7			
PH8	USB0_ID_SOC	0	
PH9	CTP_INT	0	
PH10	CTP_RST	1	
PH11	GS-INT	0	
PH12			
PH13			
PH14			
PH15			
PH16			
PH17			
PH18	LCD-BL-EN	3	
PH19			

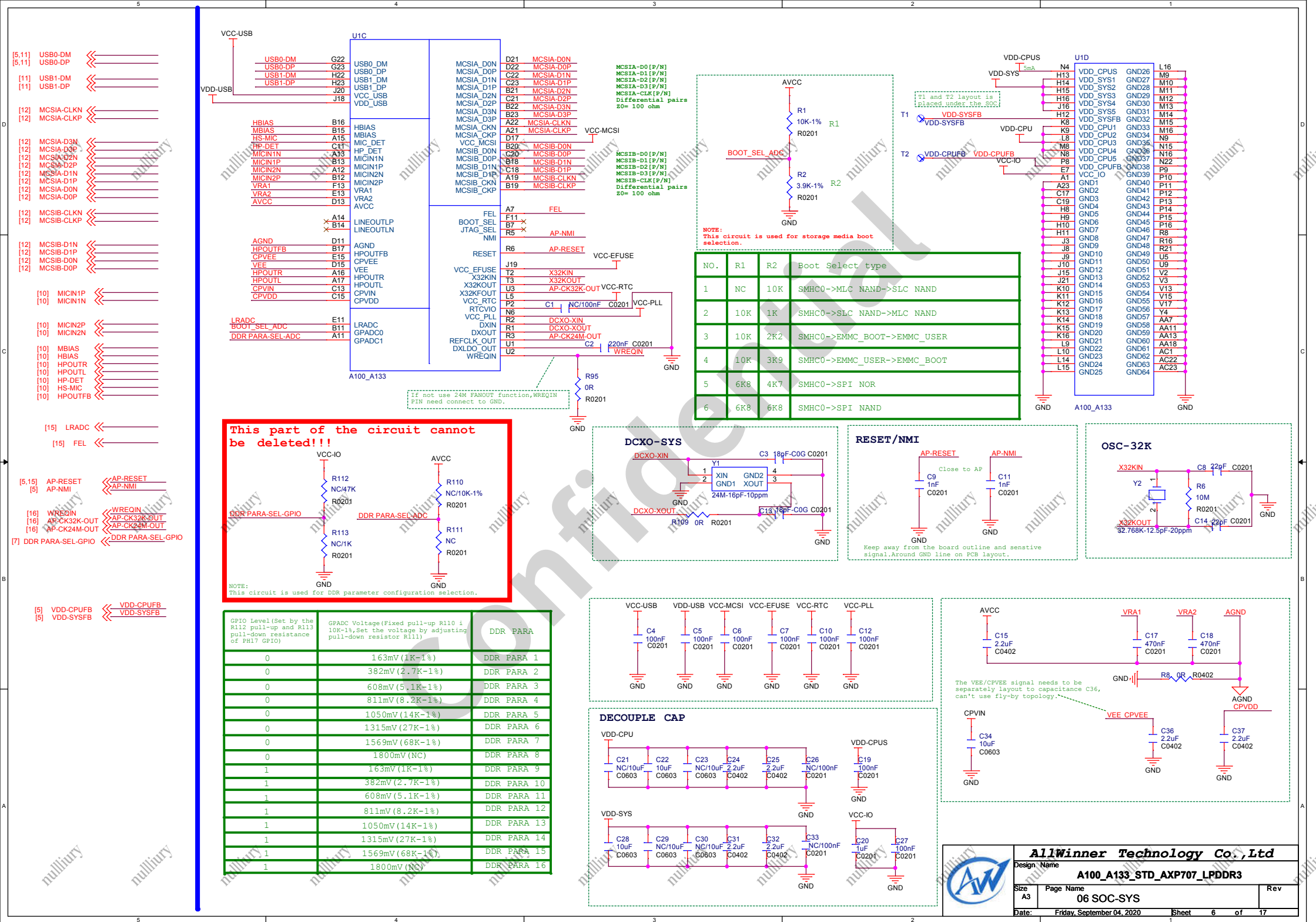
PIN	Define	CFG	Function
PC0	NAND_WE/SDC2_DS	2/3	NAND/eMMC
PC1	NAND_ALE/SDC2_RST	2/3	
PC2	NAND_CLE	2	
PC3	NAND_CE1	2	
PC4	NAND_CE0	2	
PC5	NAND_RE/SDC2_CLK	2/3	
PC6	NAND_RB0/SDC2_CMD	2/3	
PC7	NAND_RB1	2	
PC8	NAND_DQ7/SDC2_D3	2/3	
PC9	NAND_DQ6/SDC2_D4	2/3	
PC10	NAND_DQ5/SDC2_D0	2/3	
PC11	NAND_DQ4/SDC2_D5	2/3	
PC12	NAND_DQS	2	
PC13	NAND_DQ3/SDC2_D1	2/3	
PC14	NAND_DQ2/SDC2_D6	2/3	
PC15	NAND_DQ1/SDC2_D2	2/3	
PC16	NAND_DQ0/SDC2_D7	2/3	

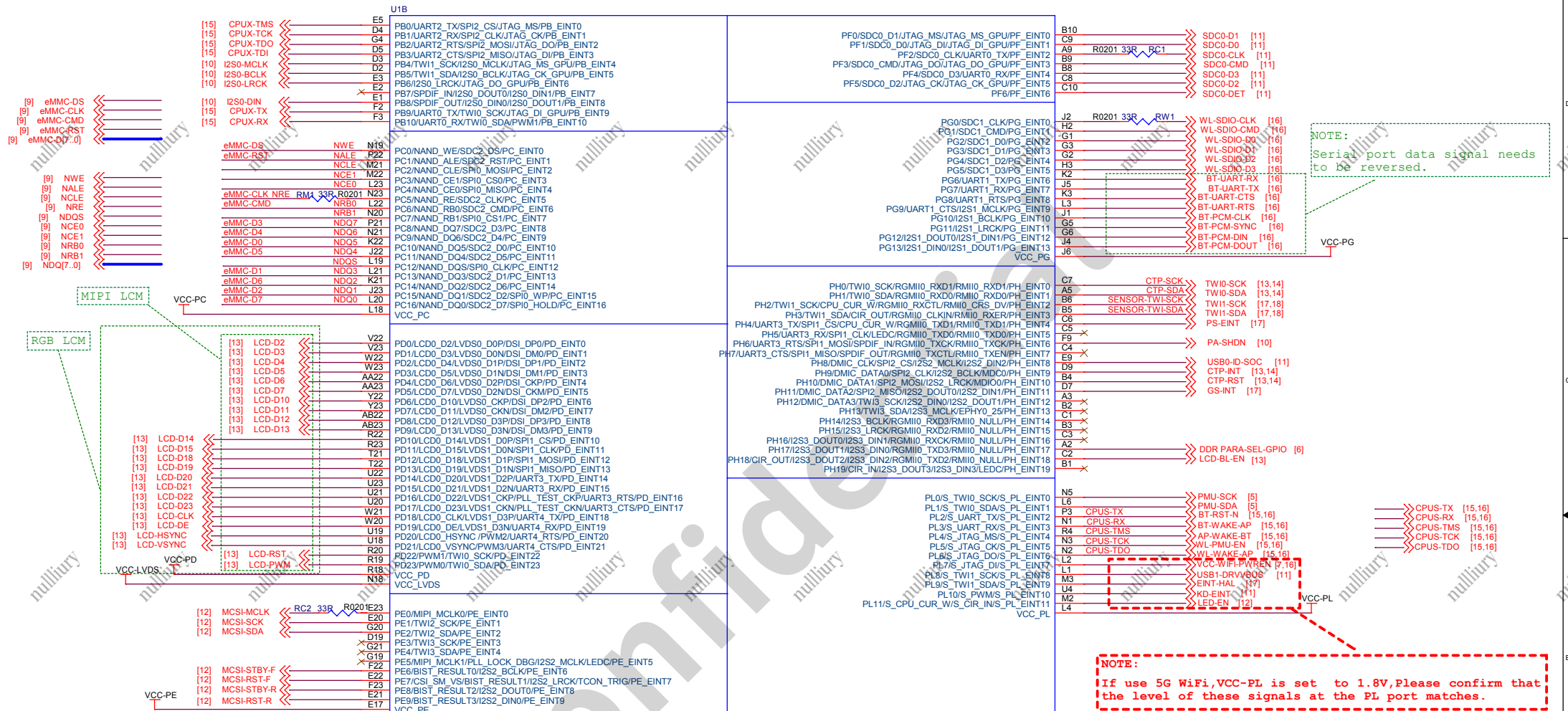
PIN	Define	CFG	Function
PF0	SDC0_D1	2	CARD
PF1	SDC0_D0	2	
PF2	SDC0_CLK	2	
PF3	SDC0_CMD	2	
PF4	SDC0_D3	2	
PF5	SDC0_D2	2	
PF6	SDC0_DET	0	

PIN	Define	CFG	Function
PL0	PMU_SCK	2	CPUS
PL1	PMU_SDA	2	
PL2	BT_RST_N	1	
PL3	BT_WAKE_AP	0	
PL4	AP_WAKE_BT	1	
PL5	WL_PMU_EN	1	
PL6	WL_WAKE_AP	0	
PL7	VCC-WIFI-PWREN	1	
PL8	USB1-DRVBUS	1	
PL9	EINT-HAL	0	
PL10	KD-EINT	0	
PL11	LED-EN	1	

PIN	Define	CFG	Function
PD0	LCD_D2	2	LCD
PD1	LCD_D3	2	
PD2	LCD_D4	2	
PD3	LCD_D5	2	
PD4	LCD_D6	2	
PD5	LCD_D7	2	
PD6	LCD_D10	2	
PD7	LCD_D11	2	
PD8	LCD_D12	2	
PD9	LCD_D13	2	
PD10	LCD_D14	2	
PD11	LCD_D15	2	
PD12	LCD_D18	2	
PD13	LCD_D19	2	
PD14	LCD_D20	2	
PD15	LCD_D21	2	
PD16	LCD_D22	2	
PD17	LCD_D23	2	
PD18	LCD_CLK	2	
PD19	LCD_DE	2	
PD20	LCD_HSYNC	2	
PD21	LCD_VSYNC	2	
PD22	LCD_RST	1	
PD23	LCD_PWM	3	

PIN	Define	CFG	Function
PG0	WL_SDIO_CLK	2	WIFI/BT
PG1	WL_SDIO_CMD	2	
PG2	WL_SDIO_D0	2	
PG3	WL_SDIO_D1	2	
PG4	WL_SDIO_D2	2	
PG5	WL_SDIO_D3	2	
PG6	BT_UART_RX	2	
PG7	BT_UART_TX	2	
PG8	BT_UART_CTS	2	
PG9	BT_UART_RTS	2	
PG10	BT_PCM_CLK	3	
PG11	BT_PCM_SYNC	3	
PG12	BT_PCM_DIN	3	
PG13	BT_PCM_DOUT	3	

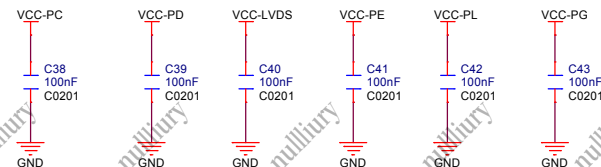




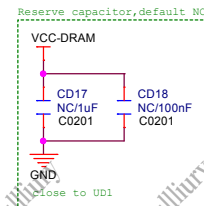
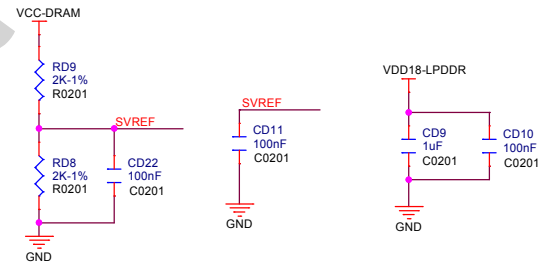
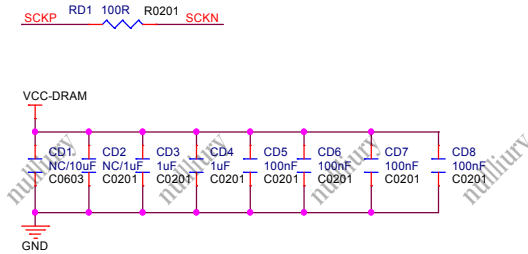
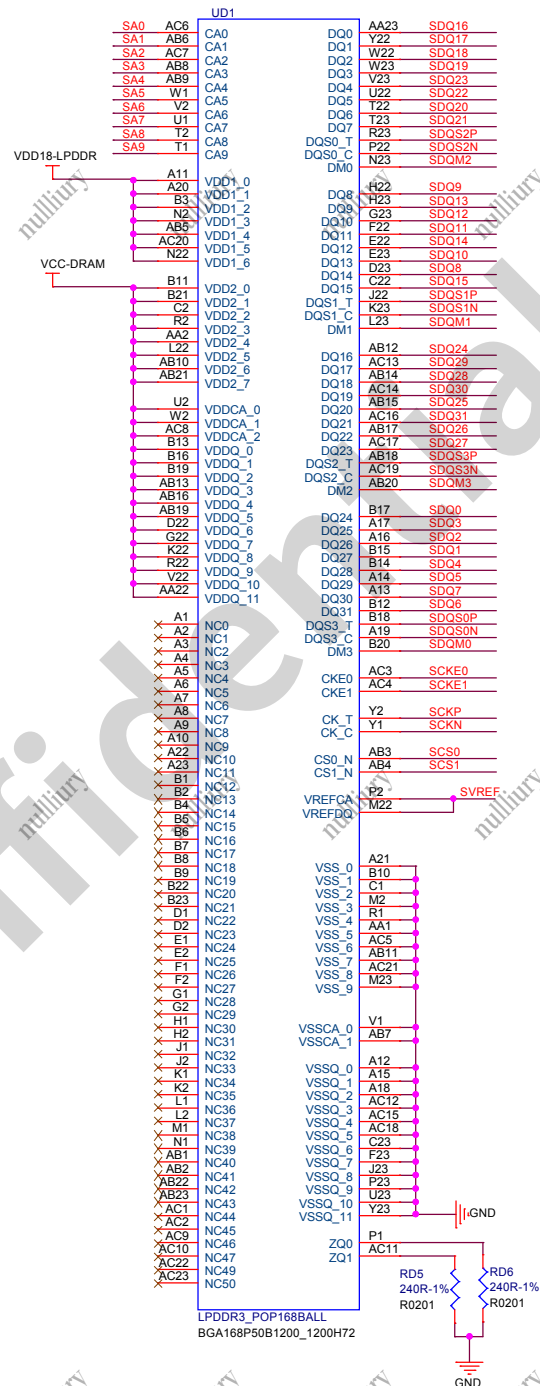
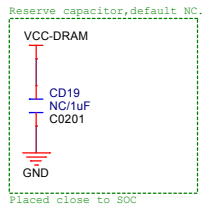
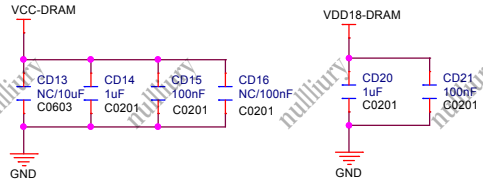
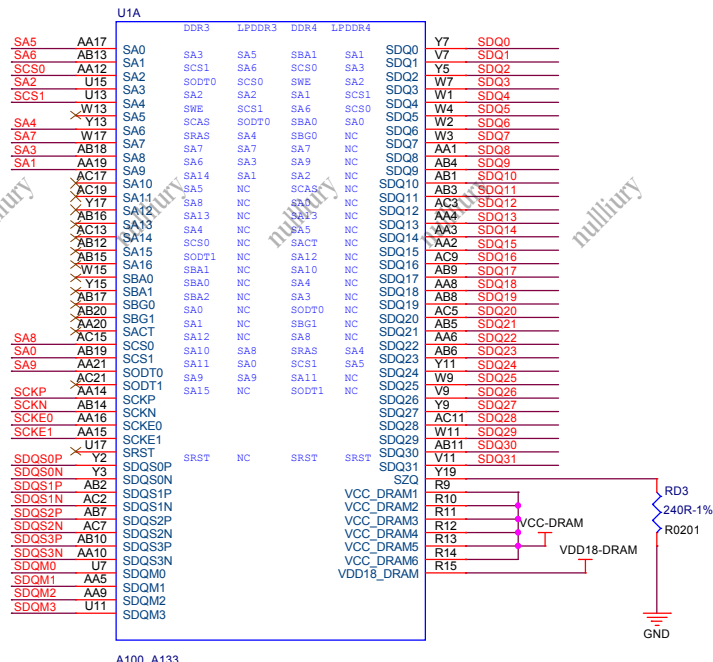
A100_A133

GPIO use guide:

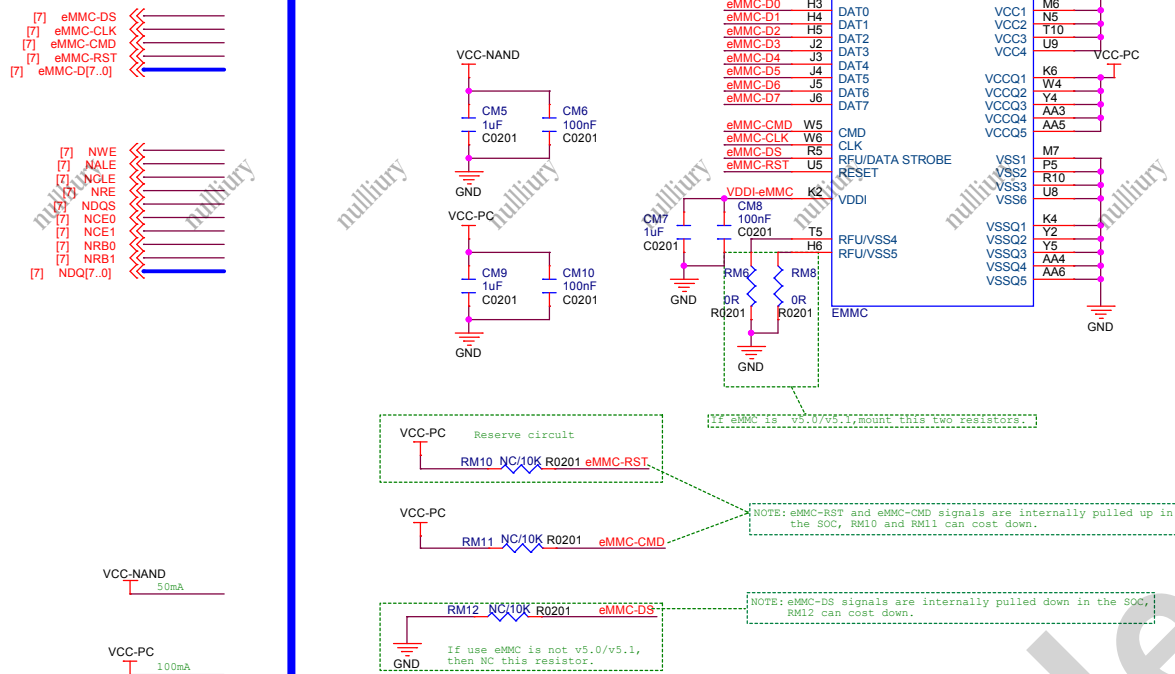
- Note that the voltage of SOC GPIO must matches the external IO voltage.
- The pull up voltage of the GPIO is selected to correspond to the power field voltage of the GPIO.



LPDDR3 32*1

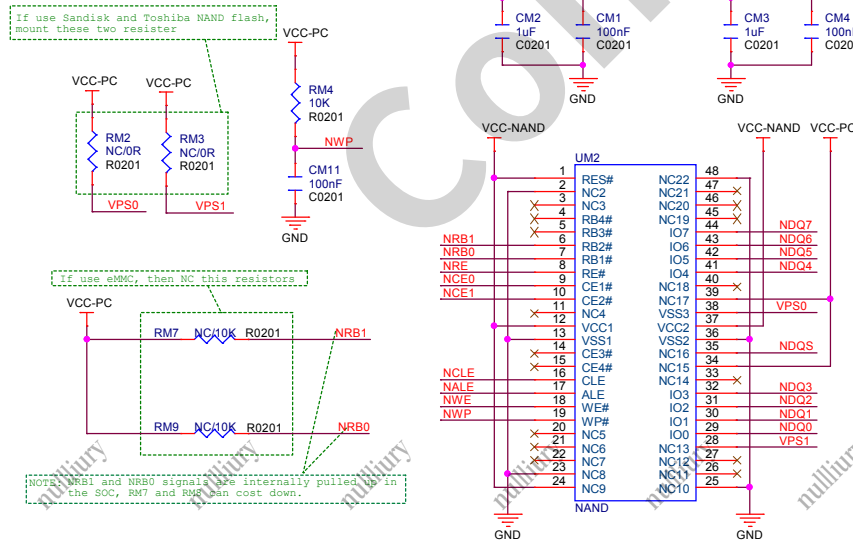


EMMC Default:use EMMC



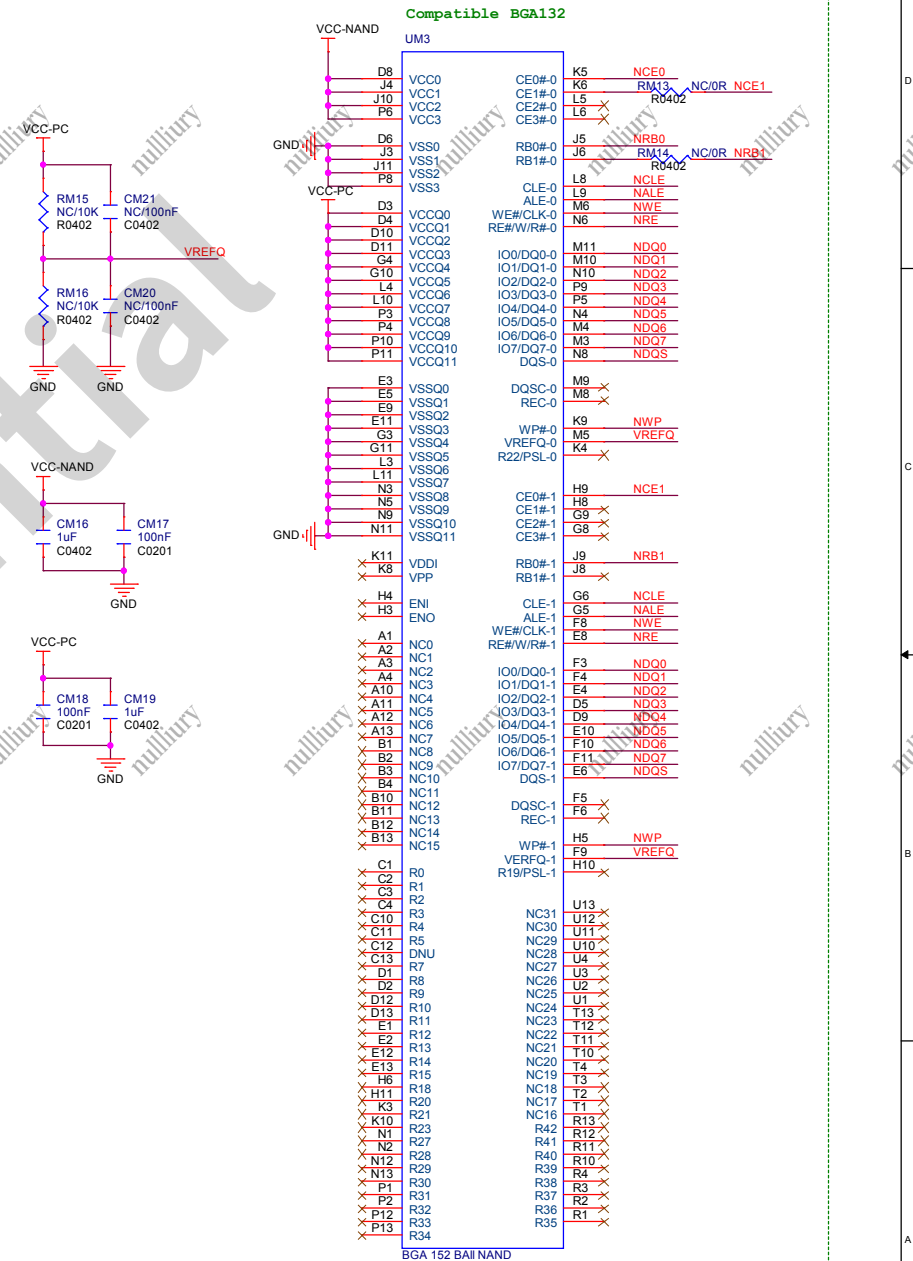
OPTION:

NAND



OPTION:

BGA NAND



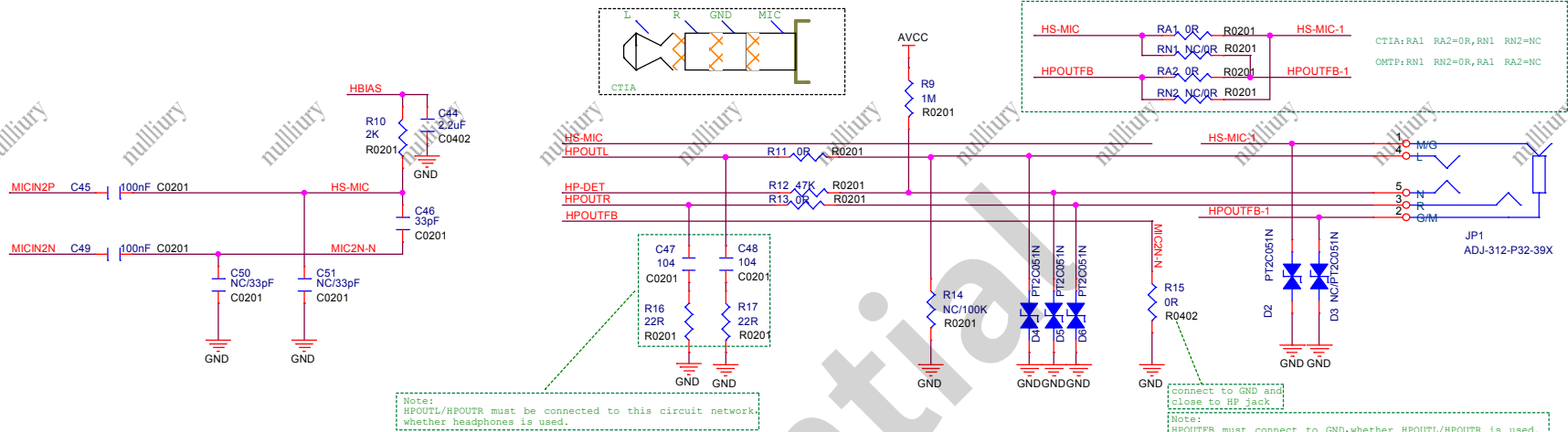
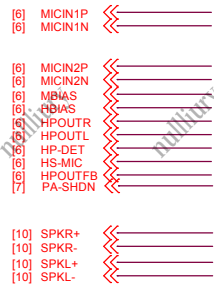
AllWinner Technology Co., Ltd

Design Name: **A100_A133_STD_AXFP07_LPDDR3**

Size: **A3** Page Name: **09 NAND/EMMC** Rev: **1**

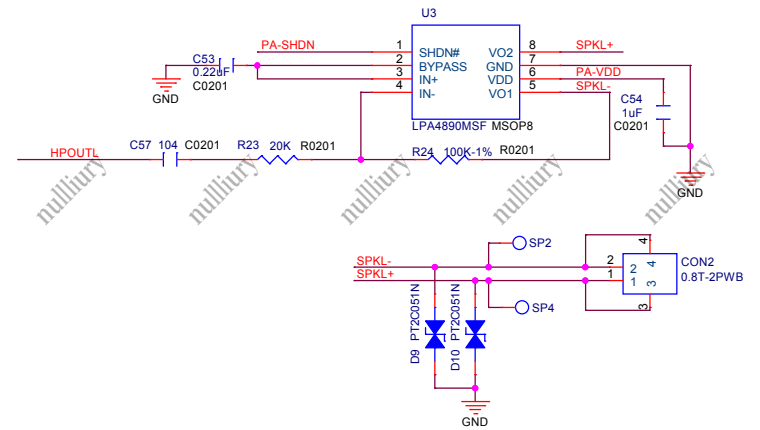
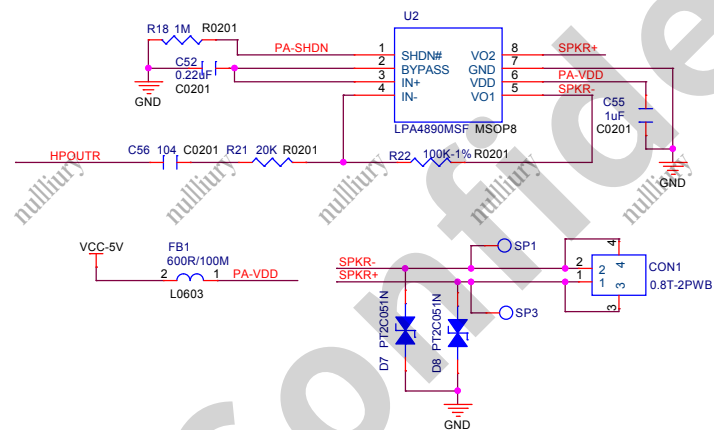
Date: **Friday, September 04, 2020** Sheet: **9** of **17**

HEADPHONE

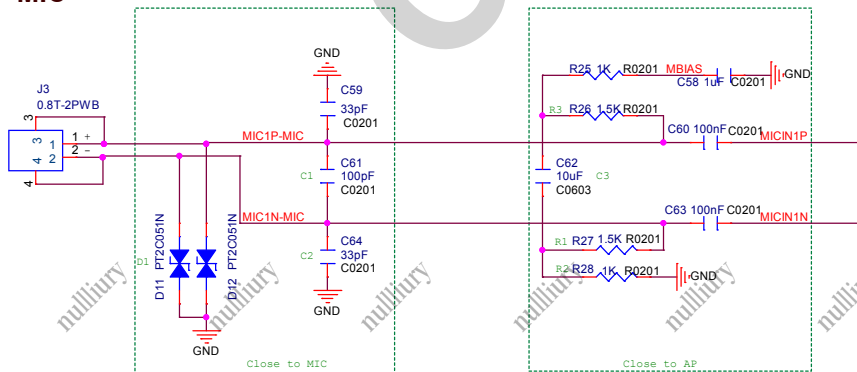


SPEAKER

Default: R-OUT, SINGLE, SPEAKER

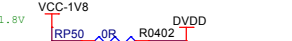


MIC

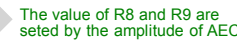


Component	Diferential	single-ended
R1 R2 C1 C3 D1	USE	NC
C2	33pF	0R
R3	1.5K	1K





I2C Address: 0x36



Component	Diferential	single-ended
R1 R2 C1 C3 D1	USE	NC
C2	33pF	0R
R3	1.5K	1K



SD CARD

NOTE: SDC0-CMD and SDC0-DET signals are internally pulled up in the SOC, RC4 and RC5 can cost down.

- [7] SDC0-D1
- [7] SDC0-D0
- [7] SDC0-CLK
- [7] SDC0-CMD
- [7] SDC0-D3
- [7] SDC0-D2
- [7] SDC0-DET

VCC-CARD
IMAX=0.8A

VCC-IO
1mA

- [5,6] USB0-DM
- [5,6] USB0-DP
- [6] USB1-DM
- [6] USB1-DP

- [7] KD-EINT

- [7] USB0-ID-SOC
- [7] USB1-DRVVBUS

VBUS

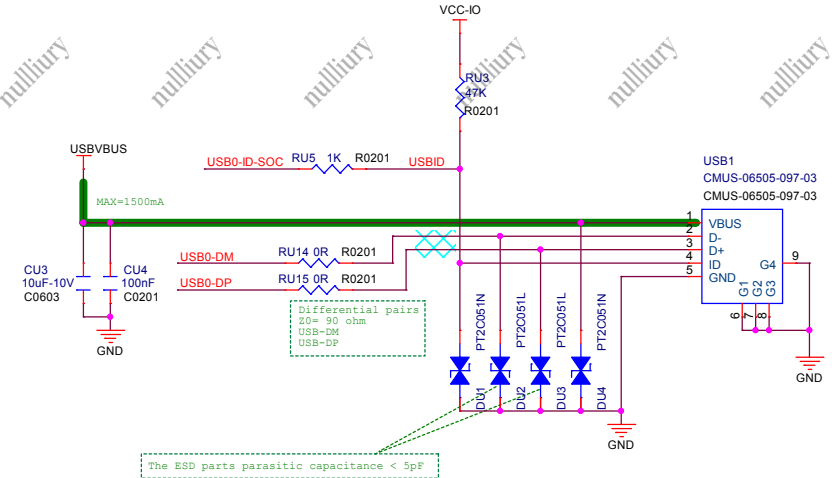
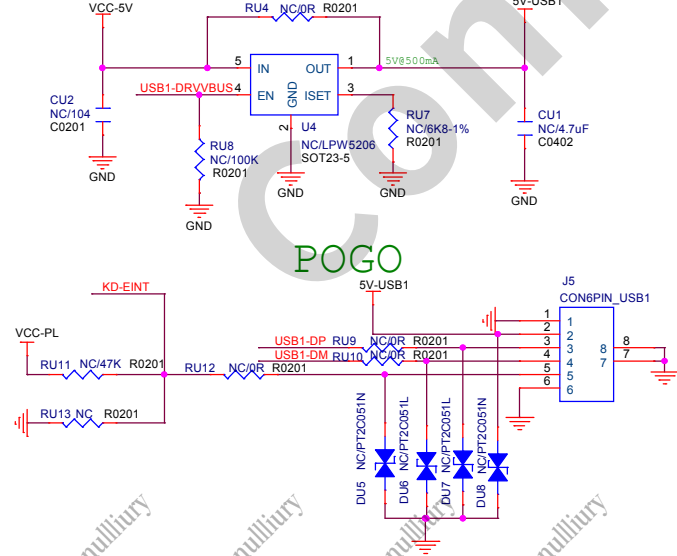
VCC-5V

VCC-PL
10mA

VCC-IO
1mA

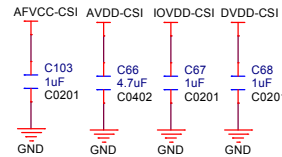
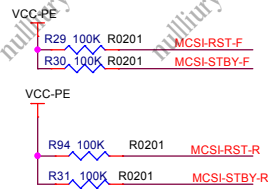
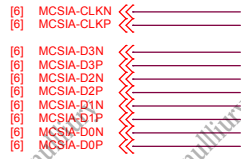
VBUS 2A USBVBUS

USB



MCSIA-D0[P/N]
MCSIA-D1[P/N]
MCSIA-D2[P/N]
MCSIA-D3[P/N]
MCSIA-CLK[P/N]
Differential pairs
Z0= 100 ohm

MCSIB-D0[P/N]
MCSIB-D1[P/N]
MCSIB-D2[P/N]
MCSIB-CLK[P/N]
Differential pairs
Z0= 100 ohm

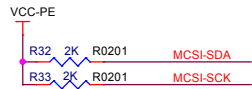


NOTE:

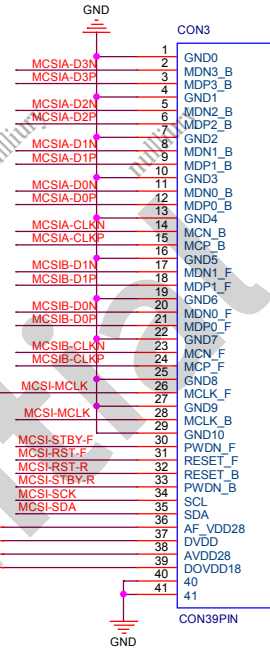
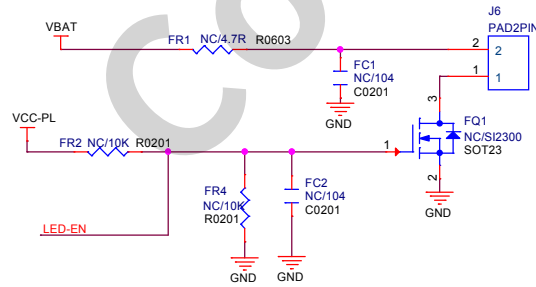
1.The working voltage and current of power need determine according to the peripheral specification.

2.AFVCC-CSI has timing requirements, don't share the same power with DVDD-CSI.

3.The front and rear cameras need to consider the compatibility of DVDD-CSI voltage.



FLASH_LED



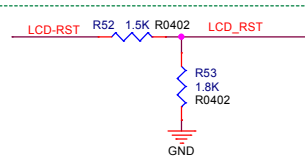
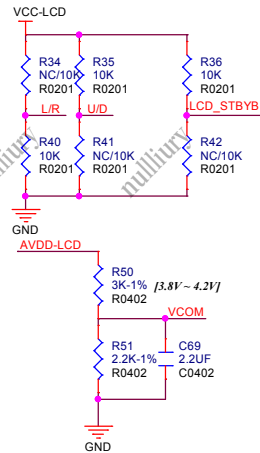
NOTE:
Close to connector



AllWinner Technology Co.,Ltd			
Design Name	A100_A133_STD_AX707_LPDDR3		
Size	Page Name	Rev	
A3	12 CAMERA		
Date:	Friday, September 04, 2020	Sheet	12 of 17

[7] LCD-D2 MIPI-DSIO-DP0
 [7] LCD-D3 MIPI-DSIO-DM0
 [7] LCD-D4 MIPI-DSIO-DM1
 [7] LCD-D5 MIPI-DSIO-CKP
 [7] LCD-D6 MIPI-DSIO-CKM
 [7] LCD-D7 MIPI-DSIO-DP2
 [7] LCD-D10 MIPI-DSIO-DM2
 [7] LCD-D11 MIPI-DSIO-DM3
 [7] LCD-D12 MIPI-DSIO-DP3
 [7] LCD-D13 LCD-RST
 [7] LCD-RST1 LCD-PWM
 [7] LCD-PWM LCD-BL-EN
 [7,14] TWIO-SCK TP-SCK
 [7,14] CTP-INT TP-INT
 [7,14] CTP-RST TP-RST

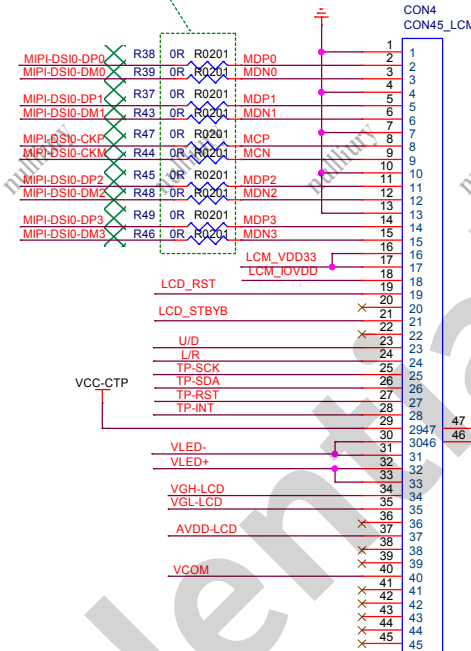
LCD



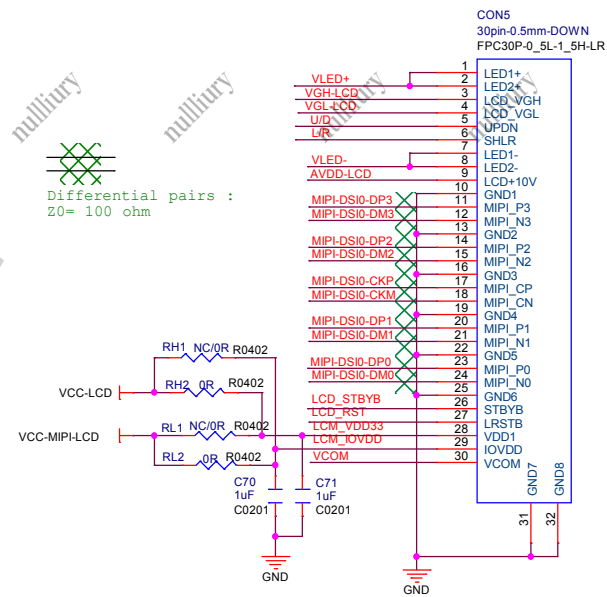
3.3V-RESET:R52=0R,R53=NC
 1.8V-RESET:R52=1.5K,R53=1.8K

Please confirm the voltage level matching of LCD-RST signal.

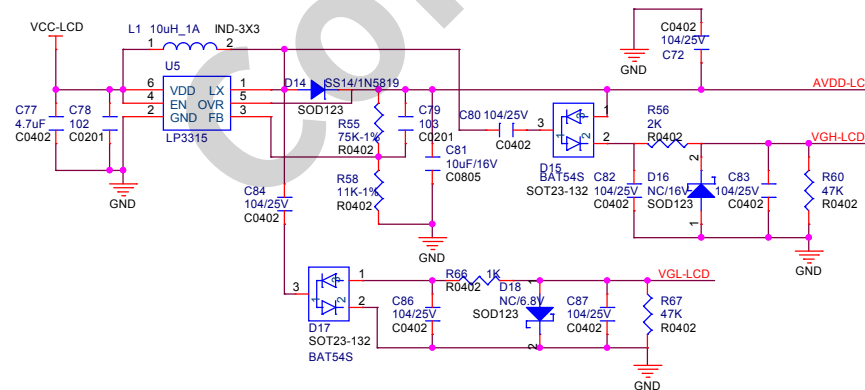
These resistors are placed under the seat to reduce PCB trace branching



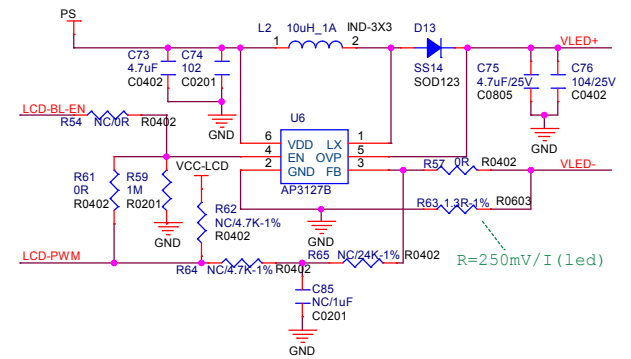
Differential pairs :
Z0= 100 ohm



LCD POWER



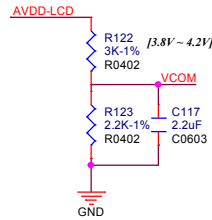
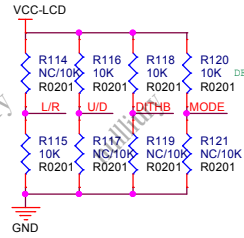
BACKLIGHT



AllWinner Technology Co.,Ltd			
Design Name	A100_A133_STD_AX707_LPDDR3		
Size	Page Name	Rev	
A3	13 LCM		
Date:	Friday, September 04, 2020	Sheet	13 of 17

- [7] LCD-D2 LCD-B2
- [7] LCD-D3 LCD-B3
- [7] LCD-D4 LCD-B4
- [7] LCD-D5 LCD-B5
- [7] LCD-D6 LCD-B6
- [7] LCD-D7 LCD-B7
- [7] LCD-D10 LCD-G2
- [7] LCD-D11 LCD-G3
- [7] LCD-D12 LCD-G4
- [7] LCD-D13 LCD-G5
- [7] LCD-D14 LCD-G6
- [7] LCD-D15 LCD-G7
- [7] LCD-D18 LCD-R2
- [7] LCD-D19 LCD-R3
- [7] LCD-D20 LCD-R4
- [7] LCD-D21 LCD-R5
- [7] LCD-D22 LCD-R6
- [7] LCD-D23 LCD-R7
- [7] LCD-CLK LCD-CLK
- [7] LCD-DE LCD-DE
- [7] LCD-HSYNC LCD-HSYNC
- [7] LCD-VSYNC LCD-VSYNC
- [7] LCD-RST LCD-RST
- [7] LCD-PWM LCD-PWM

LCD

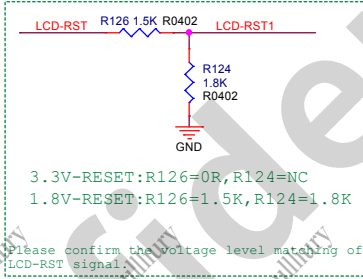


VCC-LCD

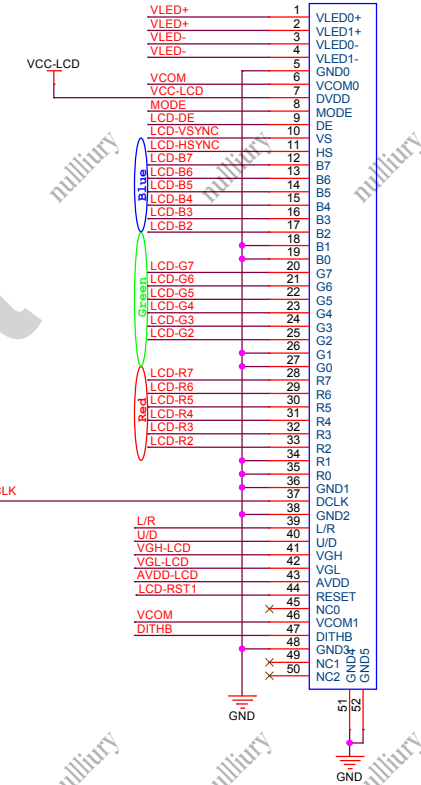
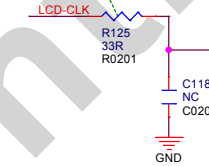
IMAX=0.3A

PS

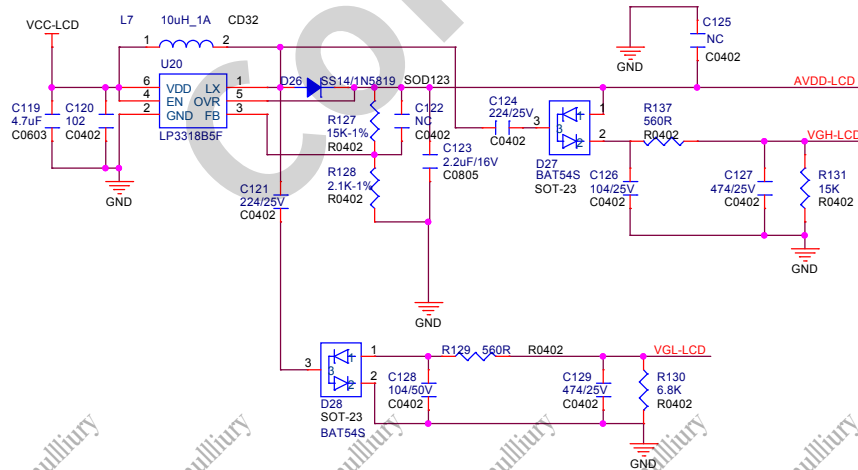
IMAX=0.5A



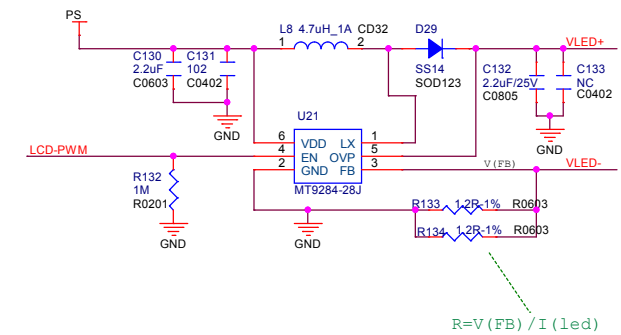
Placed close to SOC



LCD POWER

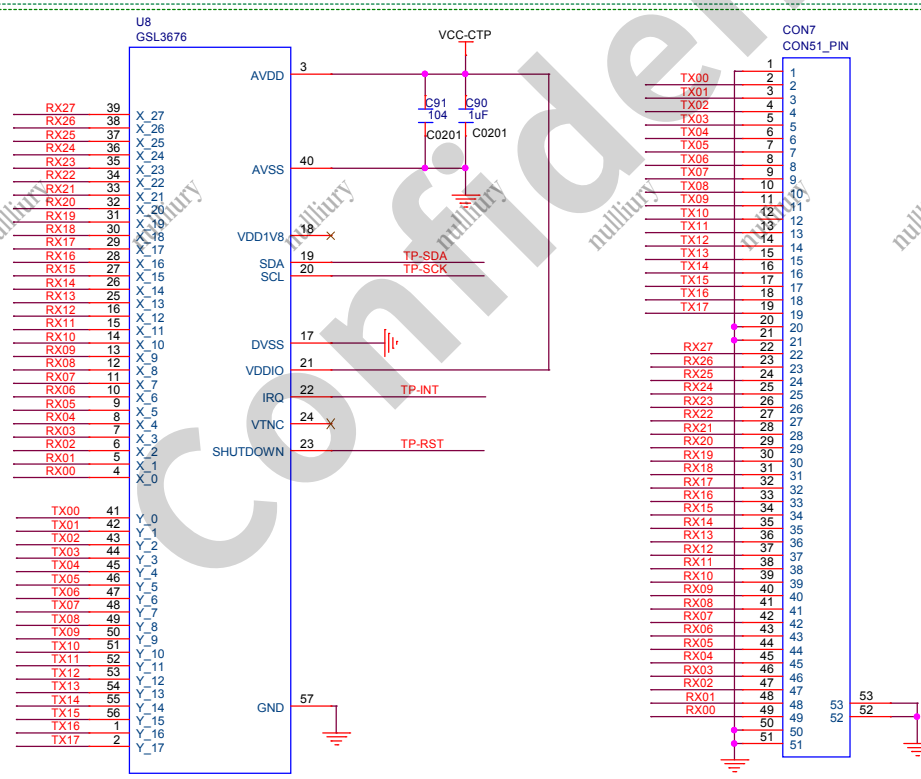
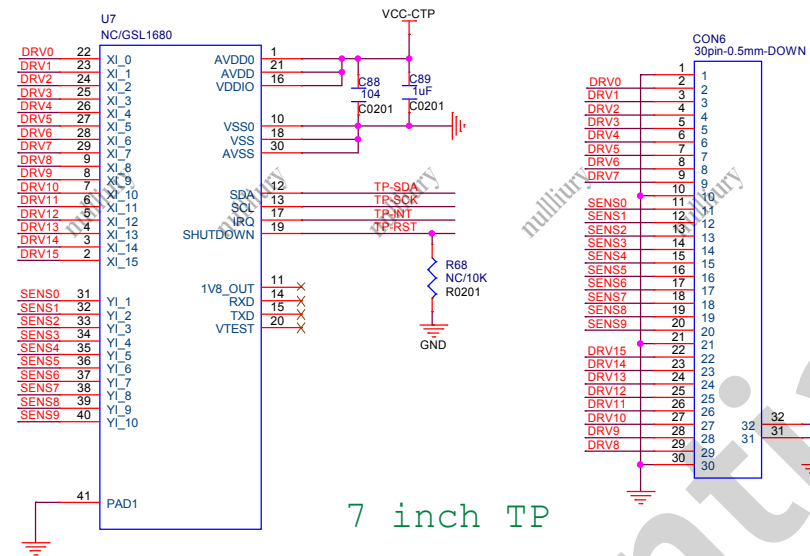
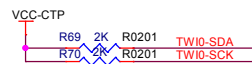


BACKLIGHT

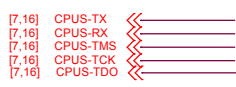
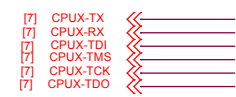
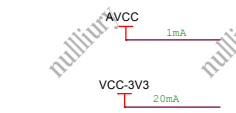


$$R = V(FB) / I(led)$$

CTP

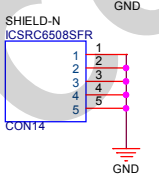
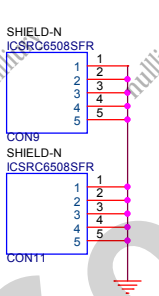
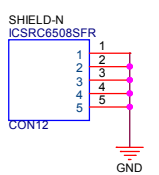
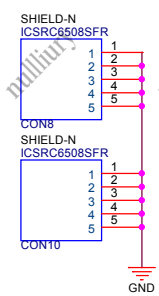


KEY

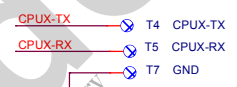


SHIELD

SOC&DDR&PMU mask clamp

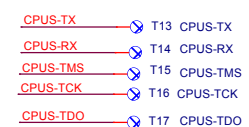


CPUX DEBUG



NOTE:Mark Signal Name Silkscreen On PCB Board

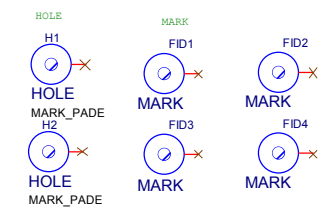
CPUS DEBUG



FEL



ASSEMBLE



AllWinner Technology Co.,Ltd			
Design Name		A100_A133_STD_AXP707_LPDDR3	
Size	Page Name	15 KEY/DEBUG	
A3		Rev	
Date: Friday, September 04, 2020		Sheet 15 of 17	

[7] WL-SDIO-CMD << WL-SDIO-CMD
[7] WL-SDIO-D0 << WL-SDIO-D0
[7] WL-SDIO-D1 << WL-SDIO-D1
[7] WL-SDIO-D2 << WL-SDIO-D2
[7] WL-SDIO-D3 << WL-SDIO-D3
[7] WL-SDIO-CLK << WL-SDIO-CLK

[7,15] WL-PMU-EN << WL-RESETN
[7,15] WL-WAKE-AP << WL-WAKE-AP

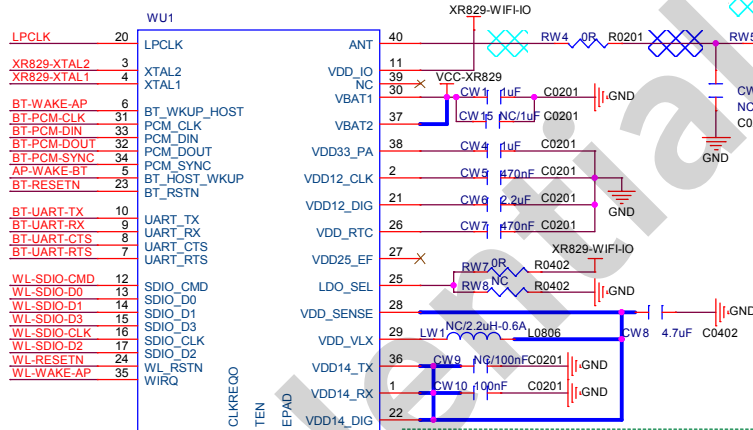
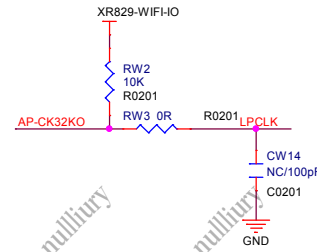
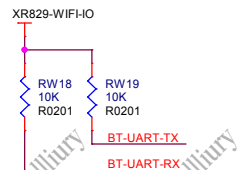
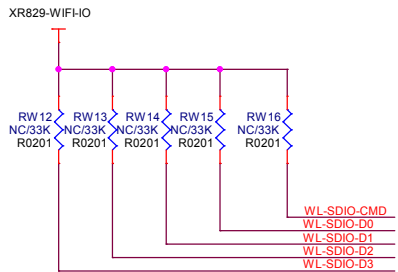
[6] AP-CK24M-OUT << XTALIN

[7,15] BT-WAKE-AP << BT-WAKE-AP
[7] BT-PCM-CLK << BT-PCM-CLK
[7] BT-PCM-DIN << BT-PCM-DIN
[7] BT-PCM-DOUT << BT-PCM-DOUT
[7] BT-PCM-SYNC << BT-PCM-SYNC
[7,15] AP-WAKE-BT << AP-WAKE-BT
[7,15] BT-RST-N << BT-RESETN

[7] BT-UART-TX << BT-UART-TX
[7] BT-UART-RX << BT-UART-RX
[7] BT-UART-RTS << BT-UART-RTS
[7] BT-UART-CTS << BT-UART-CTS

[6] AP-CK32K-OUT << AP-CK32K-OUT

2.7-5.5V 400mA VCC-WIFI VCC-XR829
3.3/1.8V 10mA VCCIO-WIFI XR829-WIFI-IO

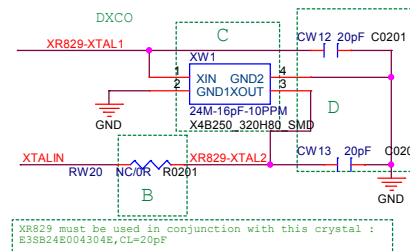


NOTE: Place capacitors near the IC



NOTE:

XR829	A	B	C	D
24M Crystal source	A	B	C	D
Crystal	NC	NC	24M-16pF-10PPM	CW12=20pF, CW13=20pF
DCXO-RFCLK	0R	0R	NC	CW12=0R, CW13=NC

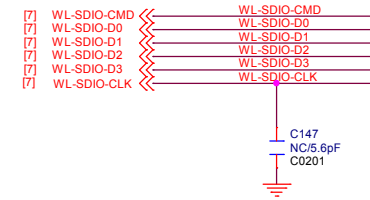
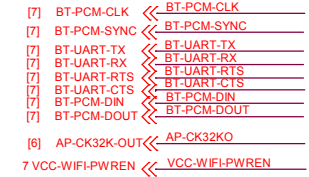
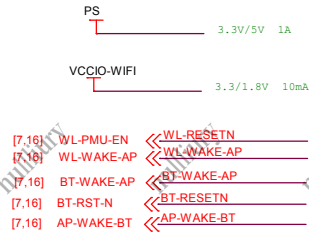


Default:use 24M crystal

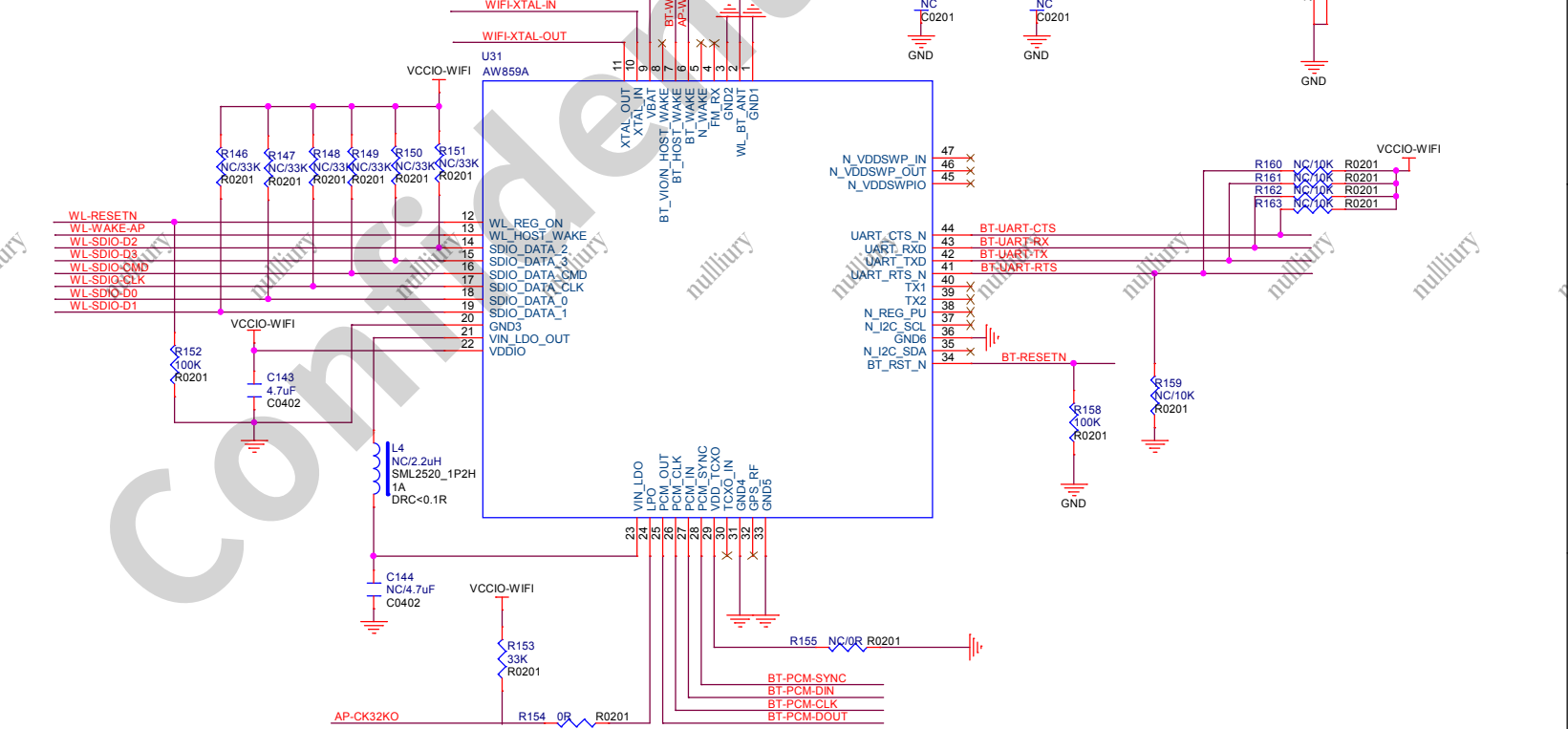
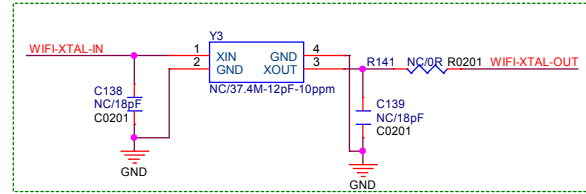
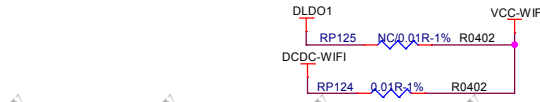
XR829 must be used in conjunction with this crystal :
E33824B004304B, CL=20pF



5G WIFI+BT



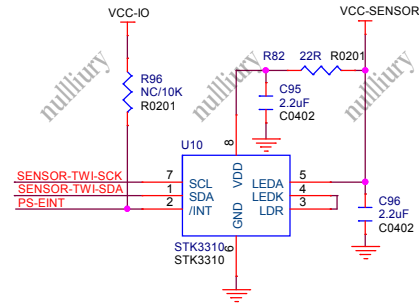
NOTE:
If use AW859A, VCCIO-WiFi and VCC -PL setting must be set to 1.8V.



ALS Sensor

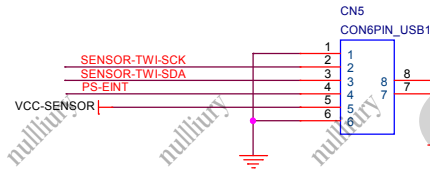
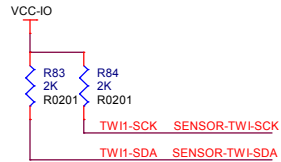
VCC-MOTOR
VCC-PL
VCC-SENSOR

[7,18] TWI1-SCK
[7,18] TWI1-SDA
[7] GS-INT
[7] EINT-HAL
[7] PS-EINT

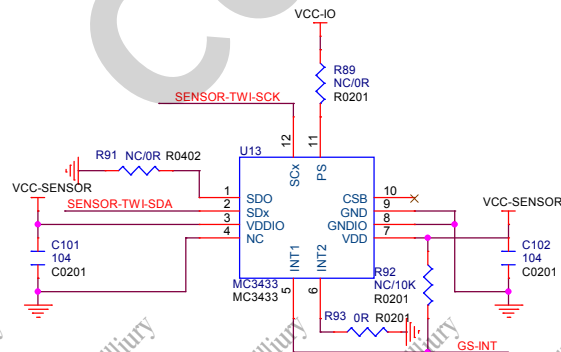


I2C Address: 0x48

ALS Sensor CON

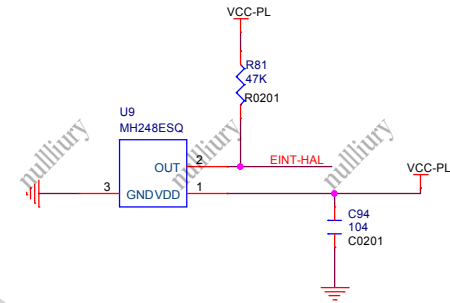


3axis G-sensor

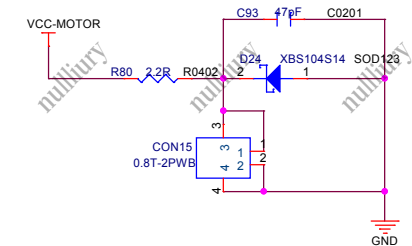


I2C Address: 0x19

Hall switch



Motor



AllWinner Technology Co., Ltd

Design Name: **A100_A133_STD_AXP707_LPDDR3**

Size: **A3** Page Name: **17 SENSOR**

Date: **Friday, September 04, 2020** Sheet **17** of **17**